EXPERIMENT 2

AIM:

To upload a CSV dataset, perform data preprocessing and statistical analysis using Pandas and visualize the sales data using Pandas and Seaborn.

CODE:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


file_path = 'sales-data-new.csv'

df = pd.read_csv(file_path)


print(df.head())


print(df.isnull().sum())


df['Sales'] = df['Sales'].fillna(df['Sales'].mean())


df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)


print(df.describe())


product_summary = df.groupby('Product').agg({

    'Sales': 'sum',
```

```
    'Quantity': 'sum'
}).reset_index()

print(product_summary)

plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)

sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()

plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.show()
```

```
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)

print(pivot_table)

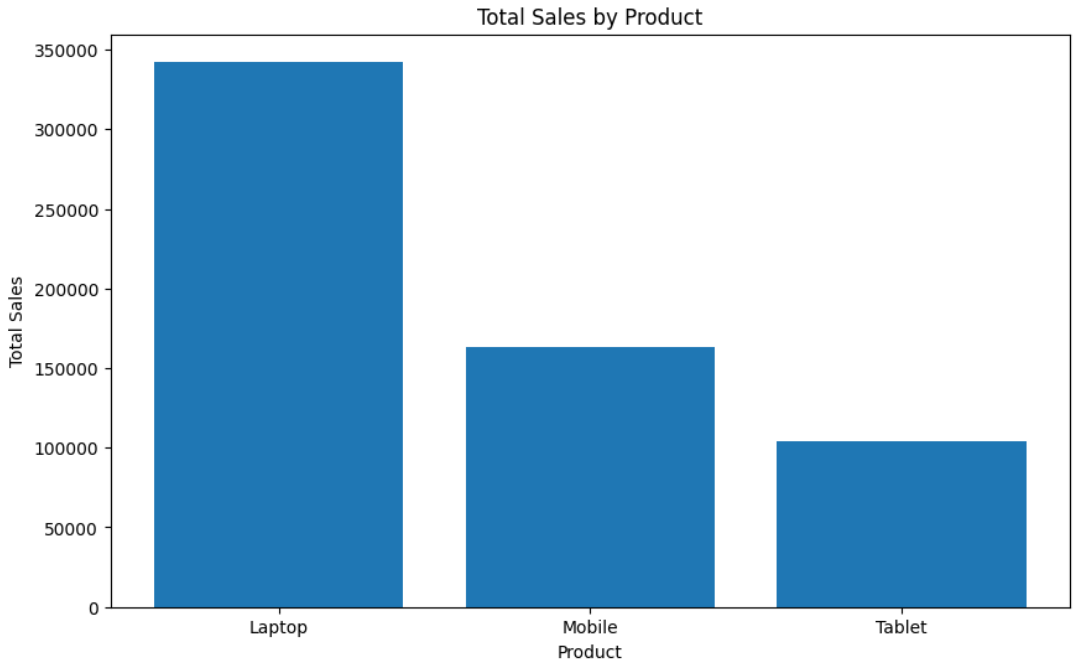
correlation_matrix = df[['Sales', 'Quantity']].corr()

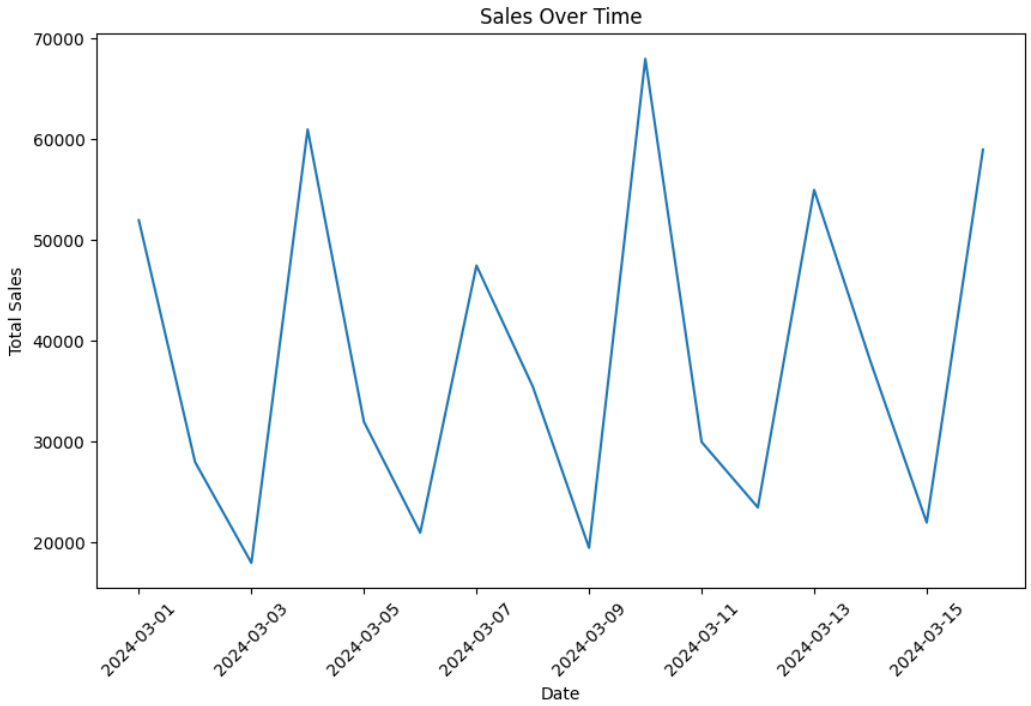
print(correlation_matrix)

plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

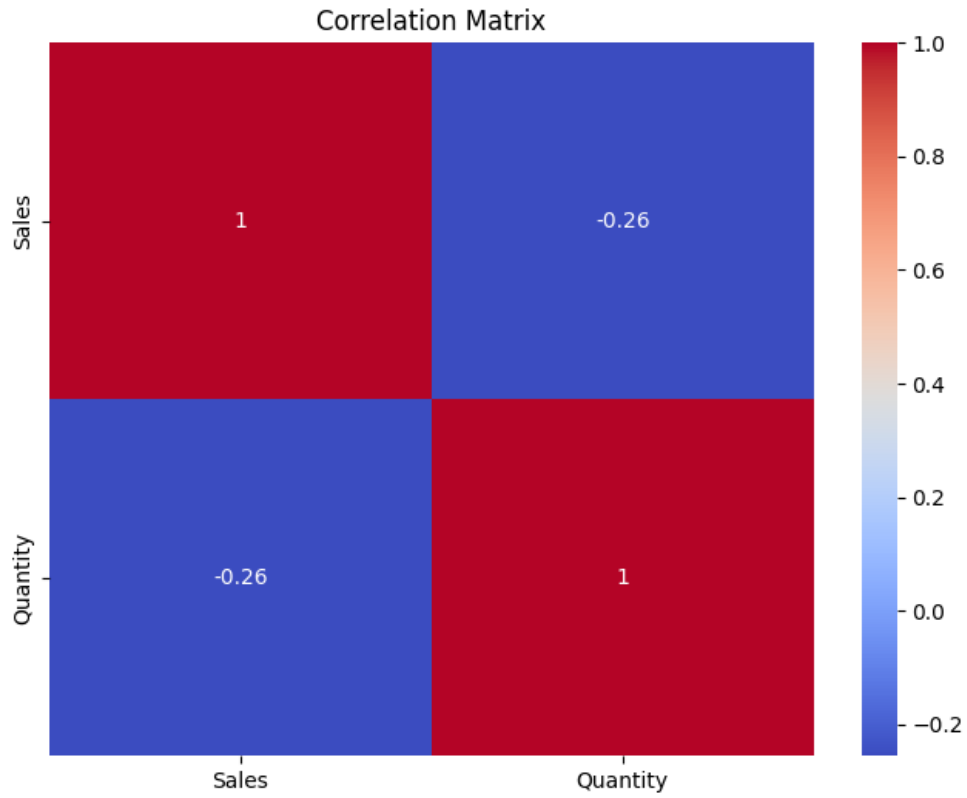
```
         Date Product  Sales  Quantity Region
0  01-03-2024  Laptop  52000         4  North
1  02-03-2024  Mobile  28000         7  South
2  03-03-2024  Tablet  18000         5   East
3  04-03-2024  Laptop  61000         5   West
4  05-03-2024  Mobile  32000         8  South
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
              Sales   Quantity
count     16.000000  16.000000
mean   38125.000000   5.750000
std    16630.794729   1.693123
min    18000.000000   3.000000
25%    23125.000000   4.750000
50%    33750.000000   5.500000
75%    52750.000000   7.000000
max    68000.000000   9.000000
  Product  Sales  Quantity
0  Laptop 342500        27
1  Mobile 163500        39
2  Tablet 104000        26
```





Product	Laptop	Mobile	Tablet
Region			
East	0	38000	58500
North	222500	0	0
South	0	90000	22000
West	120000	35500	23500

	Sales	Quantity
Sales	1.0000	-0.2557
Quantity	-0.2557	1.0000



RESULT:

The dataset was successfully loaded and analysed using Pandas. The preprocessing operation was performed successfully and the imputation of missing values were present.